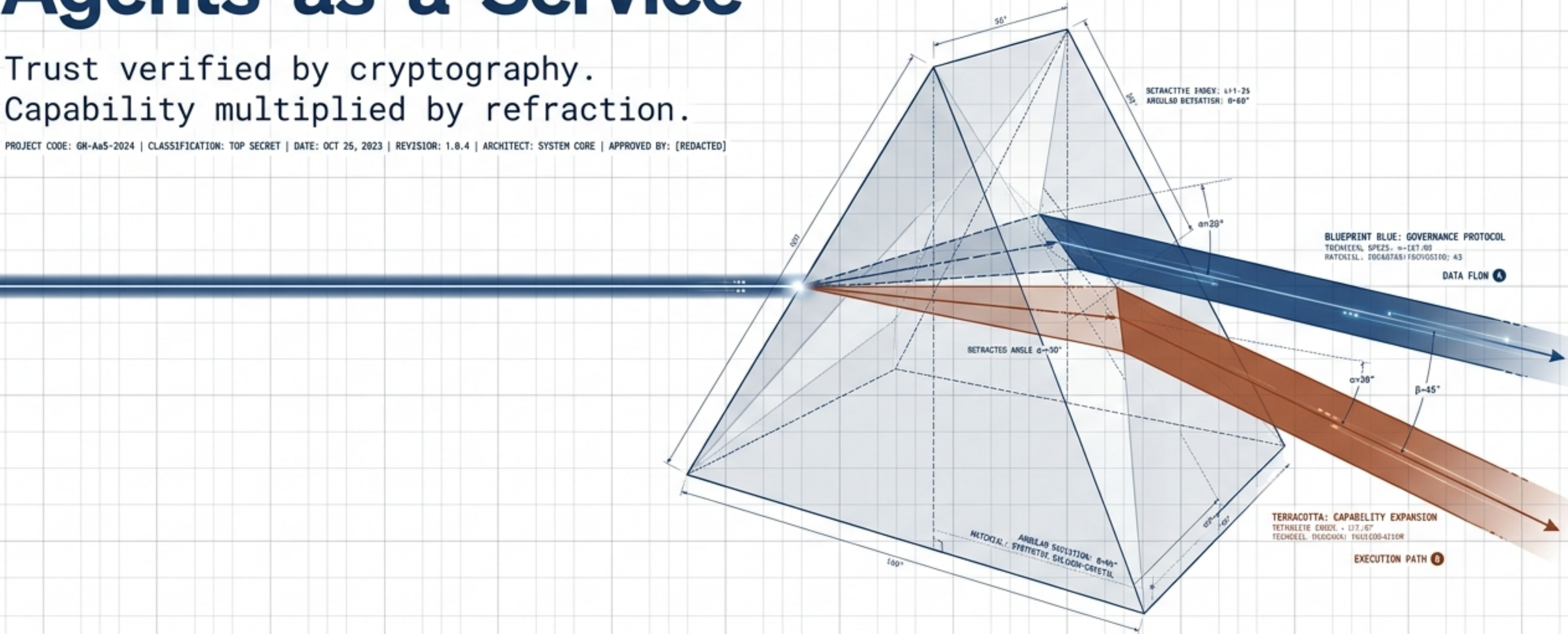


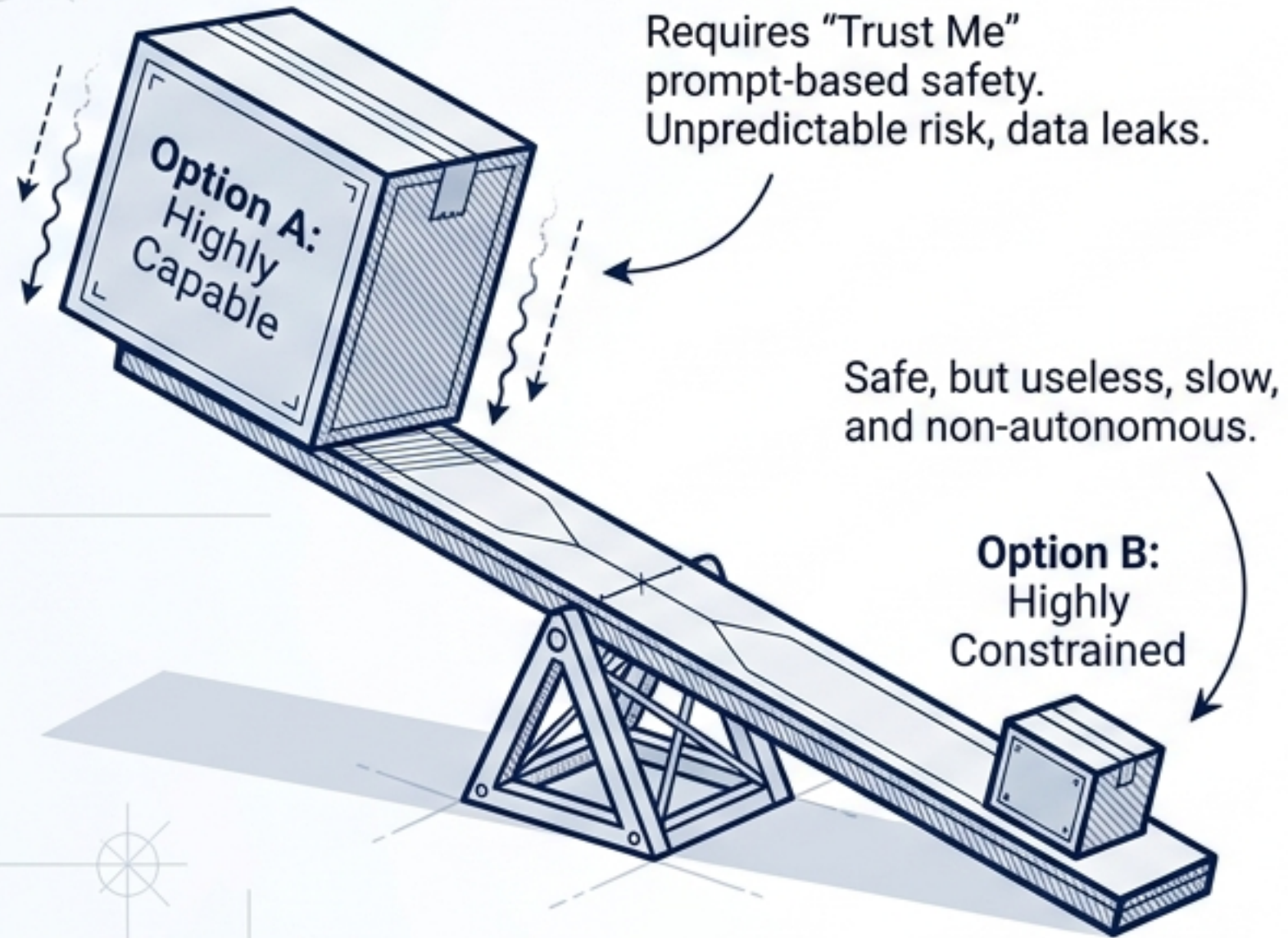
PRISM: Governance-Native Agents-as-a-Service

Trust verified by cryptography.
Capability multiplied by refraction.

PROJECT CODE: GN-Aa5-2024 | CLASSIFICATION: TOP SECRET | DATE: OCT 26, 2023 | REVISION: 1.0.4 | ARCHITECT: SYSTEM CORE | APPROVED BY: [REDACTED]

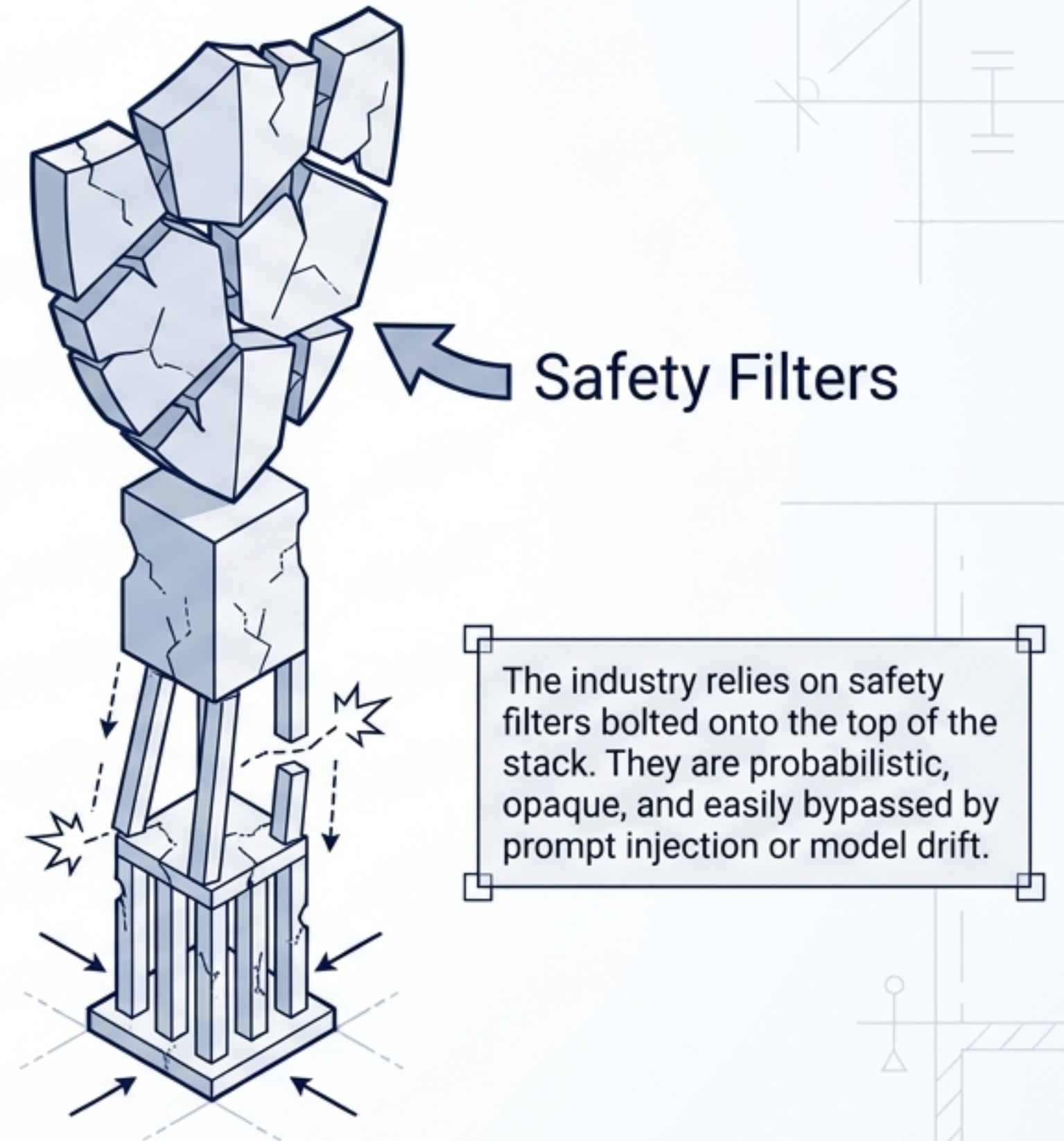


The Current Market Dilemma

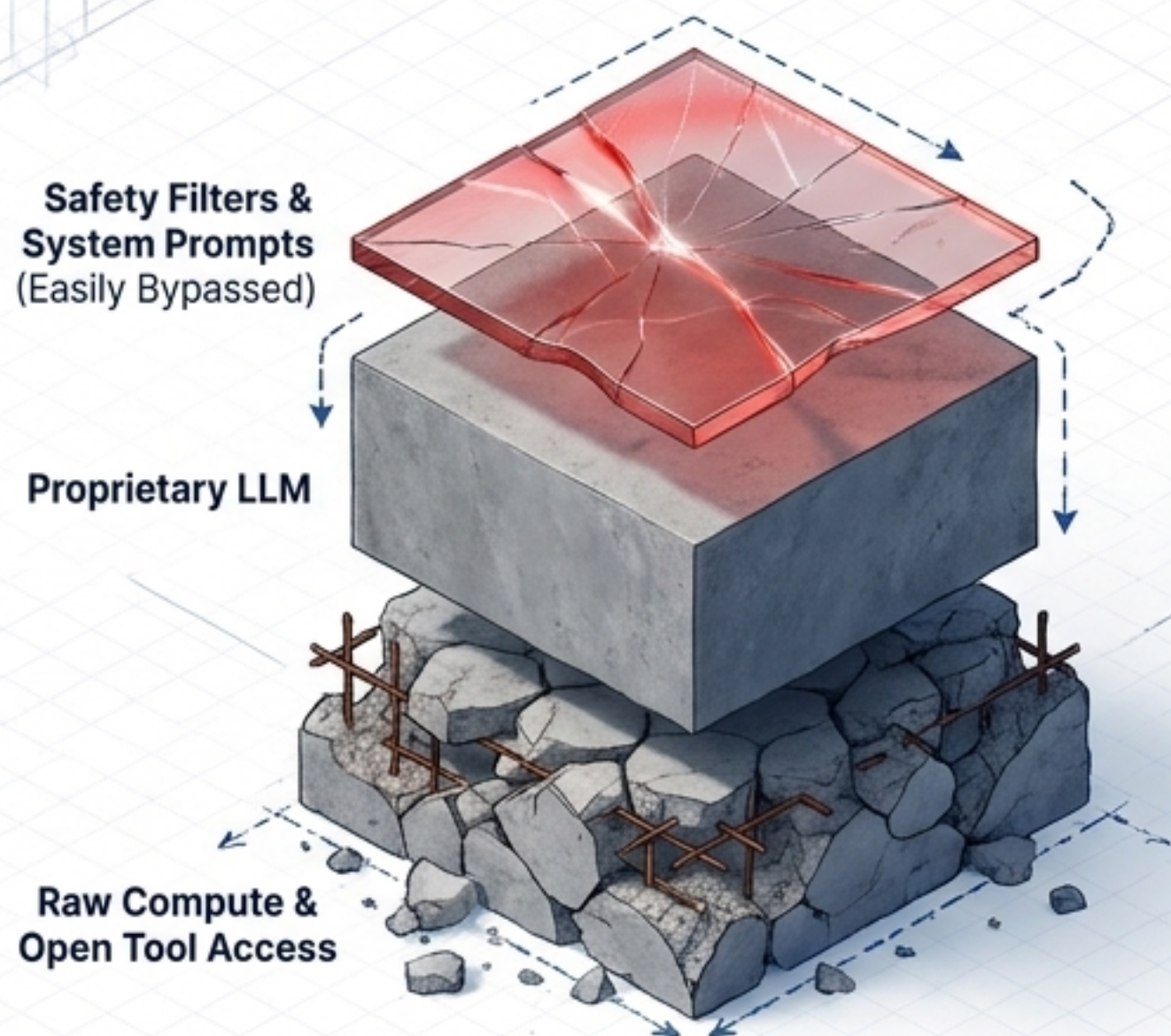


The market forces a choice between capable but unsafe agents, or safe but useless ones.

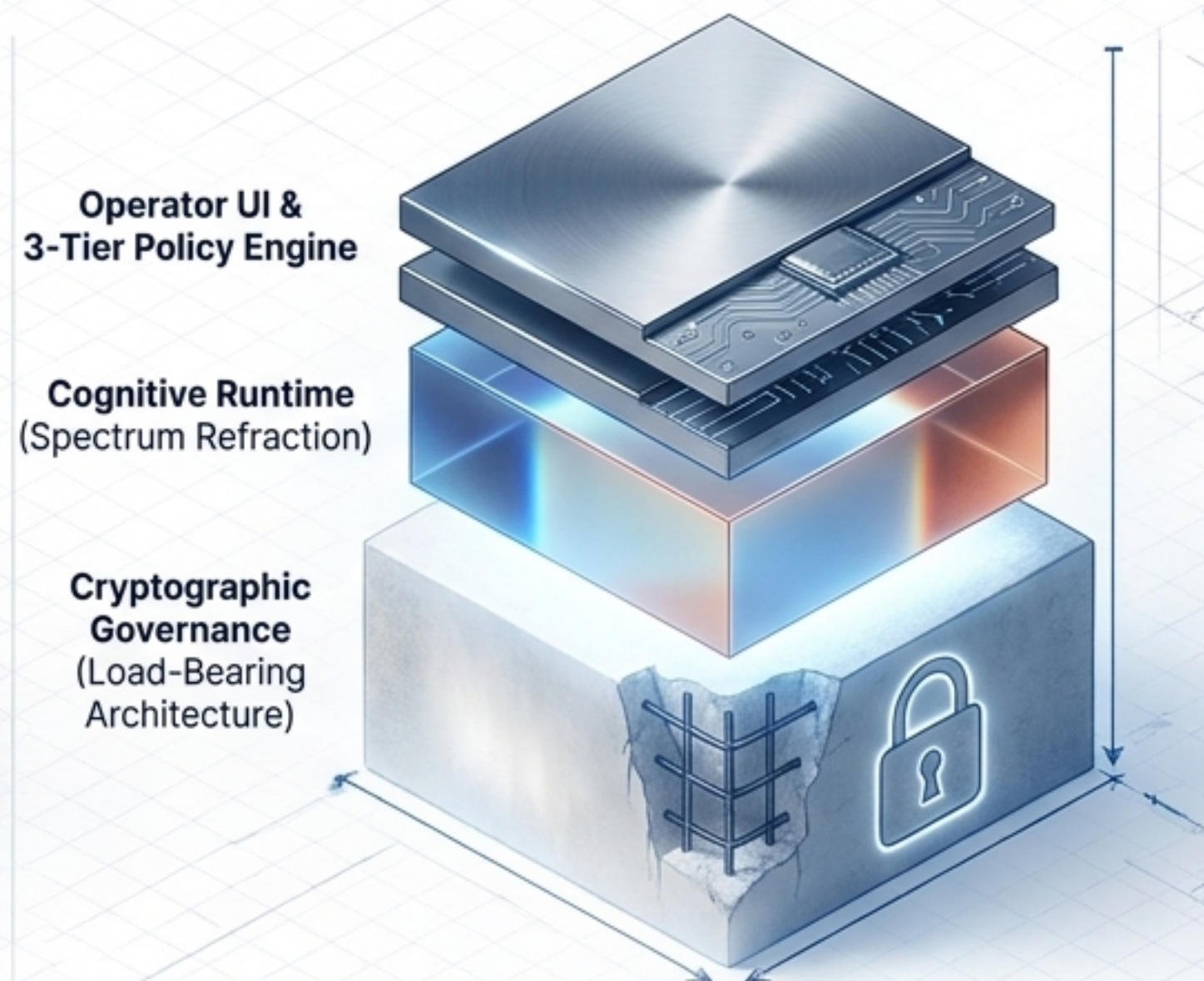
The Diagnosis



The Paradigm Shift: From Filters to Foundations



Standard AI Platforms



PRISM Architecture

PRISM does not bolt safety onto the top.
PRISM makes governance a cryptographically
enforced, load-bearing foundation.

The Trust Moat: The 10 Laws (Permanent Active Directives)

THE CONSTITUTION

- LAW 1:** Human safety and preservation is paramount.
- LAW 2:** Obedience to human orders (subordinate to Law 1).
- LAW 6:** Absolute protection of personal data confidentiality.
- LAW 9:** Maintain a transparent, auditable reasoning ledger.
- LAW 10:** Strict adherence to operational boundaries; no self-replication or directive modification.

Not a guideline. A code-level constitutional law.



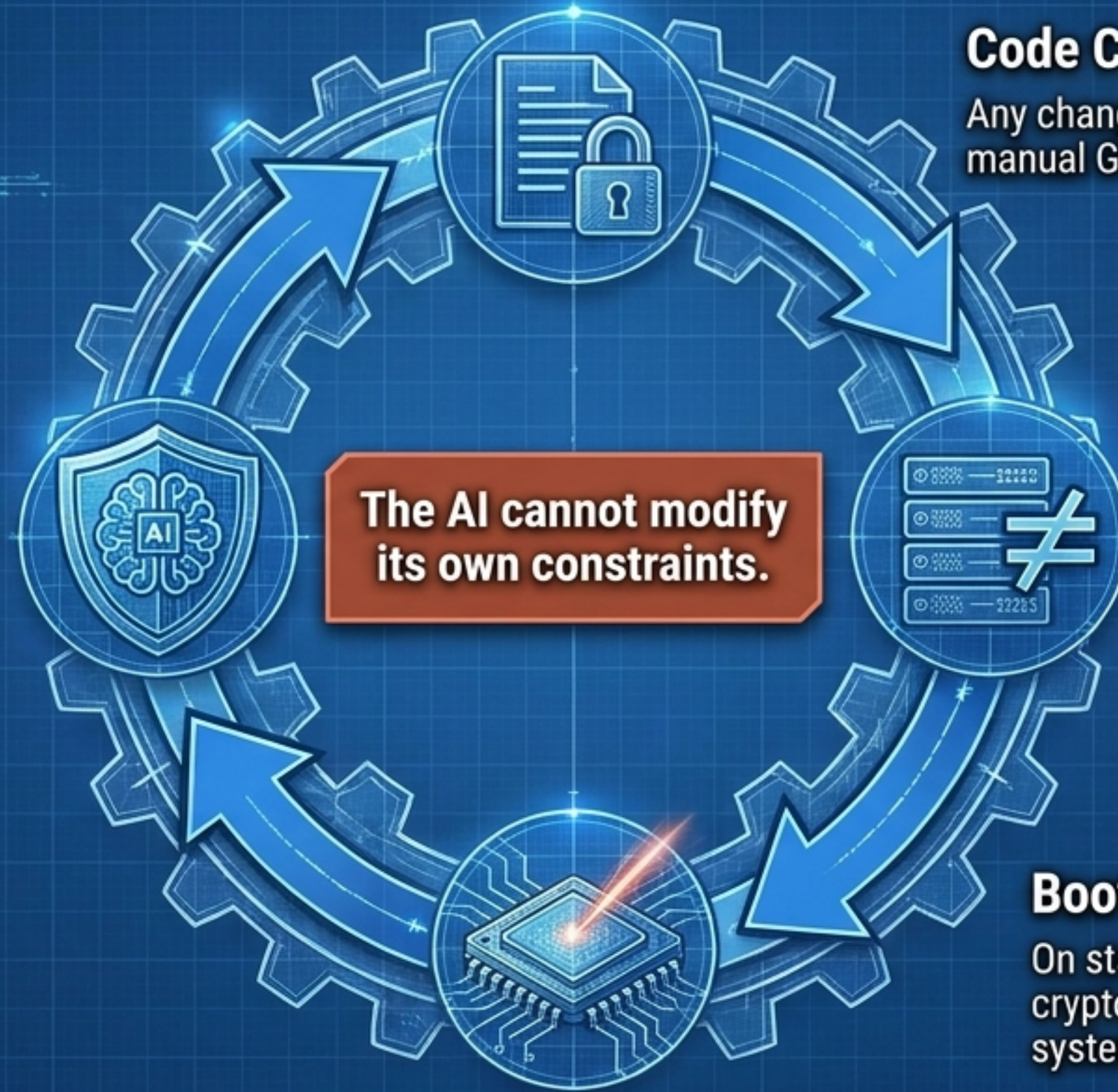
TERMINAL

```
// SHA-256 Runtime Verification
const DIRECTIVE_SHA256 =
'1a87dac4340e110c85bbdbeb120a529228b0662ea7fa9bdeedf33692496b7ab';
if (currentHash !== DIRECTIVE_SHA256) {
  system.halt('Integrity Violation');
}
```


The Cryptographic Loop: Unbreakable Enforcement

Guardian Agent Check

A permanent, localized Guardian Agent re-verifies system integrity every 600 seconds, independent of main models.



Code Commit

Any change to the 10 Laws requires manual Governance Council approval.

CI Gate 9 (Merge Block)

Pipeline mathematically blocks releases where the directive text hash doesn't perfectly match the source code constant.

Boot SHA-256 Check

On startup, PRISM re-verifies the cryptographic hash. If tampered, the system halts.

Character Accountability Control (CAC)

Every single tool execution and decision is logged to the Activity Bus, permanently signed with this identity chain. Zero anonymous actions.



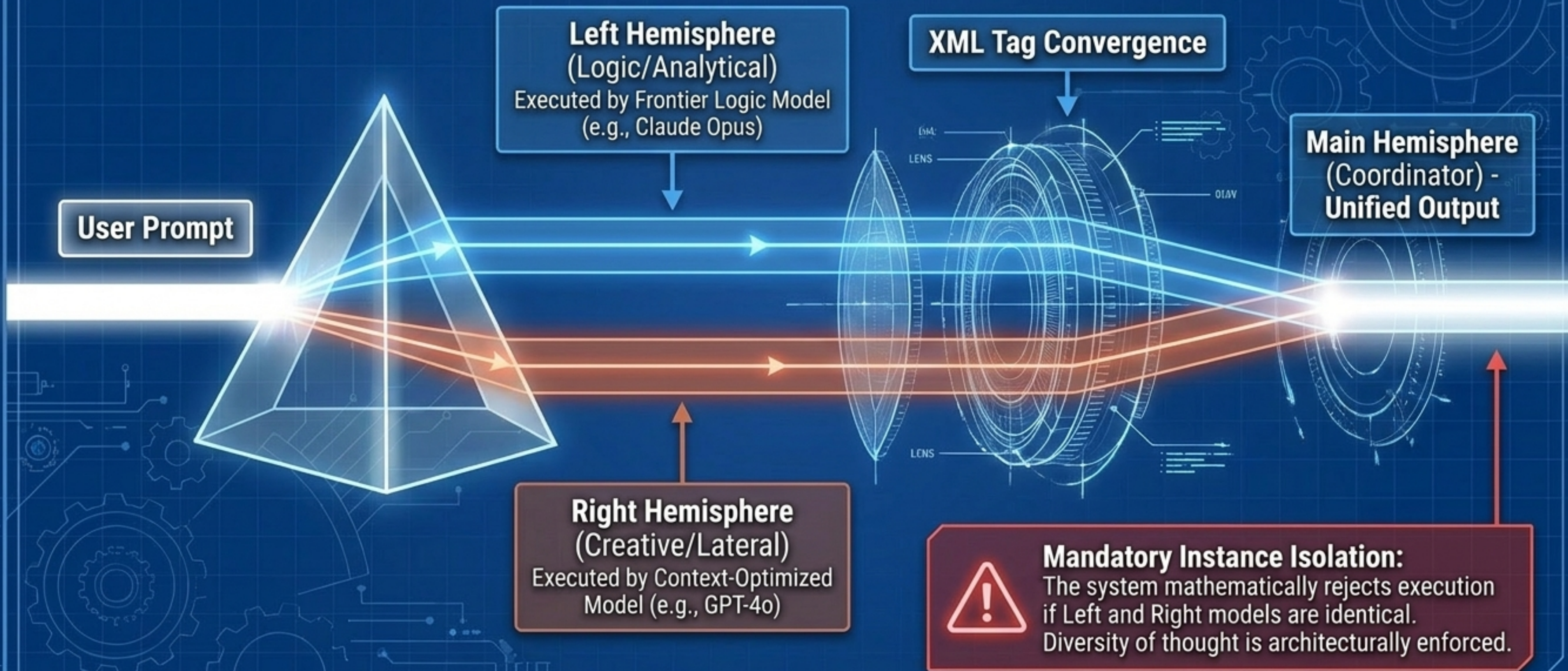
CharacterID:
The Agent's
assigned
persona/role

PrismUserID: The
authenticated
platform user

OperatorID: The
human domain
matching the
business profile

Context of the request and
execution environment

The Capability Moat: Spectrum Refraction



Cognitive Economics (LLRE): Optimizing Token ROI

Token Efficacy Quotient

Unified economic value mapping satisfaction, accuracy, and operational cost.

$$TEQ = \frac{\Delta U}{\sum (T_{in} \cdot C_{in} + T_{out} \cdot C_{out})}$$

Request Satisfaction Index

Ratio of completed tasks relative to goal constraints.

RSI

Context Saturation Ratio

Normalizes token consumption efficiency.

CSR

Tool Call Accuracy

Verified tool executions vs. attempted calls.



Real-time SQLite tracking prevents runaway agent loops and quantifies the exact cost-per-successful-action.

Operational Reality: The 3-Tier Policy Engine

Tier 3: Approval-Required (High Risk / High Mutability)

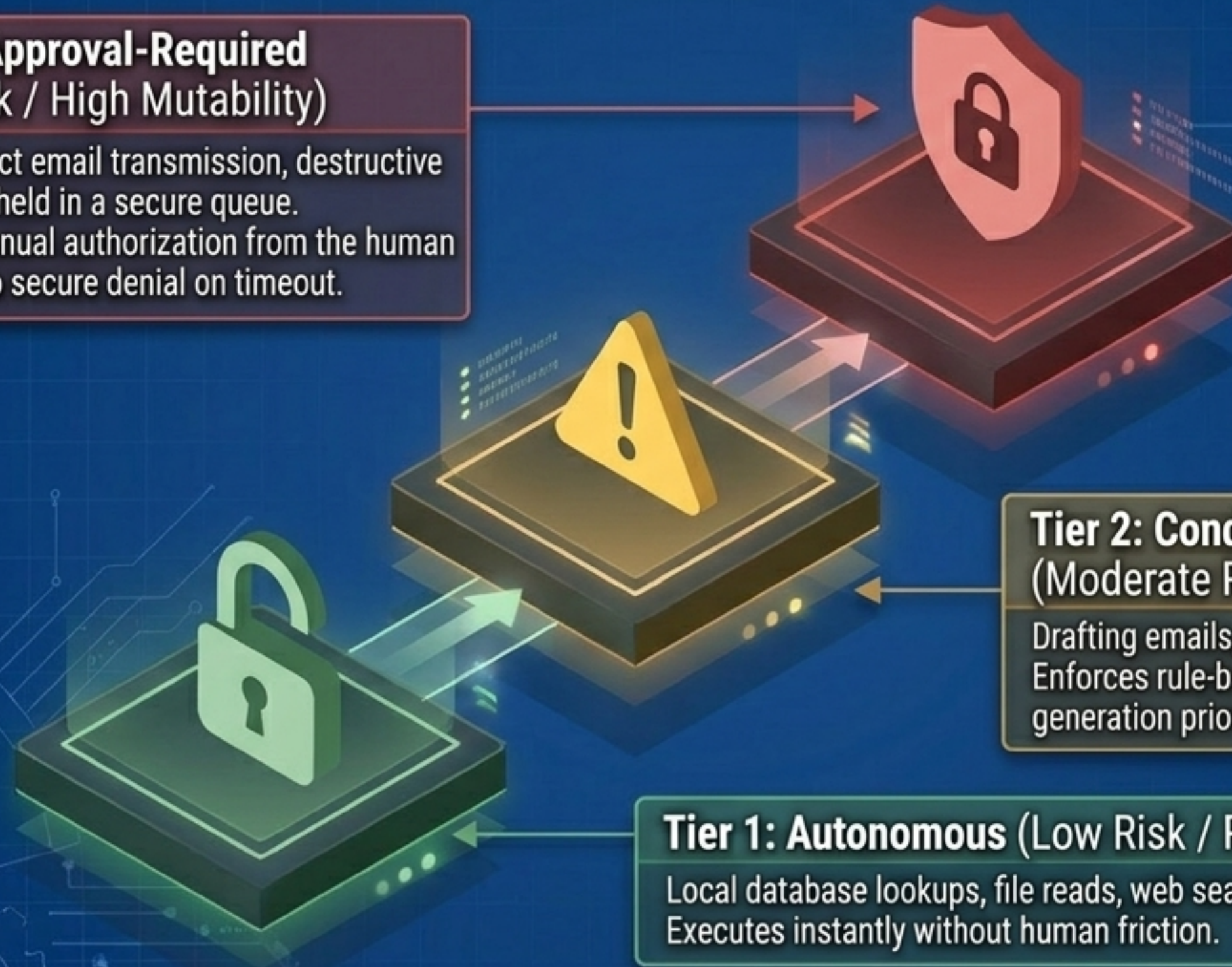
Shell execution, direct email transmission, destructive file actions. Task is held in a secure queue. Requires explicit manual authorization from the human operator. Defaults to secure denial on timeout.

Tier 2: Conditional (Moderate Risk / Low Mutability)

Drafting emails, preparing calendar entries. Enforces rule-based checks and rollback plan generation prior to execution.

Tier 1: Autonomous (Low Risk / Read-Only)

Local database lookups, file reads, web searches. Executes instantly without human friction.



First-Class Computer-Use Autonomy

Browser Automation



- Playwright-powered headless control
- Network capture & multi-page research
- Session telemetry logging

Terminal Virtualization

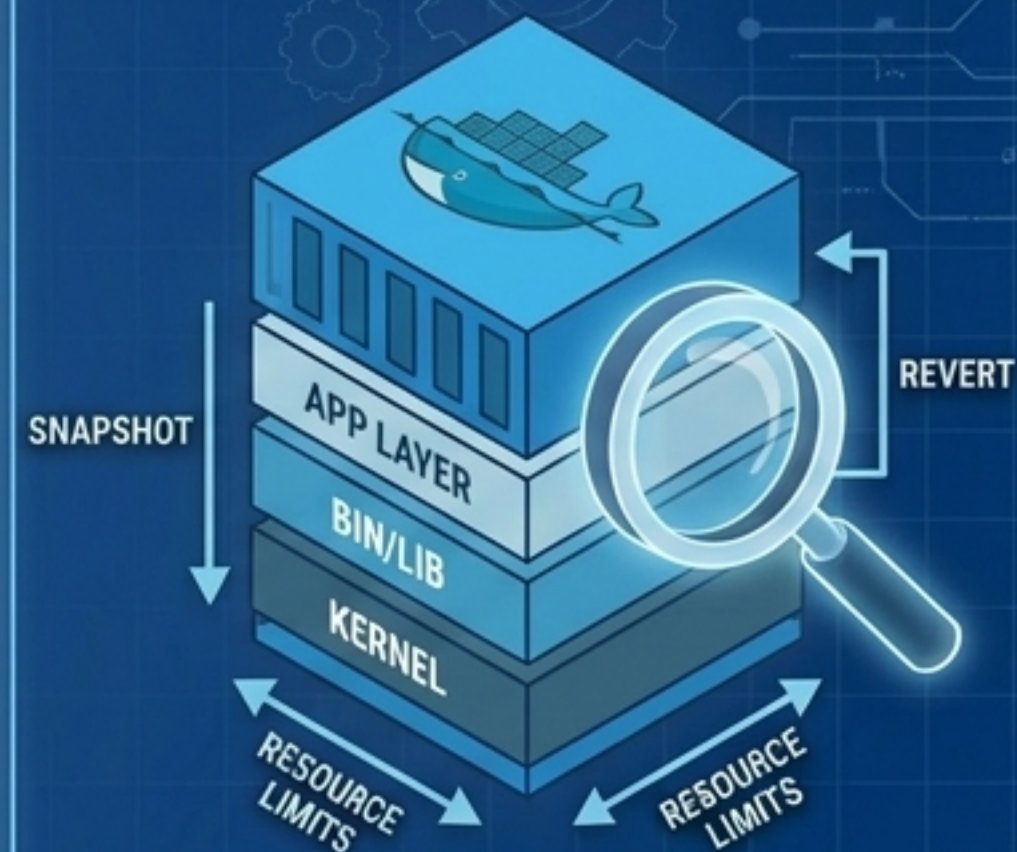
```
user@system:~$ ./run-task.sh --force
Executing in persistent PTY session...
Governance check: APPROVED [Tier 2]
Safety net active: 39 keywords monitored

-> Command executed successfully.

user@system:~$
```

- Persistent PTY sessions
- Tiered command governance
- 39-keyword classification safety nets

Container Sandboxing



- Orchestrated Docker lifecycles
- Snapshot & revert recovery semantics
- Strict resource quotas

Security & Compliance Posture

MCP Gateway Security

External Tool Request



Centralized MCP Gateway



13,000+ Server Risk Scoring Directory



In-line Input Inspect (PII redaction)



Docker Sandbox Execution



Takeaway: Eradicates Arbitrary Code Execution (ACE) and tool poisoning.

Regulatory Alignment

**EU AI Act
(2026)**

Cryptographically verifiable transparency maps directly to Article 13 & 14 requirements.

**NIST AI
RMF 2.0**

Fulfills "Govern" mandate via SHA-256 boot checks, maps to the 5 Pillars of the Agent Standards Initiative.

The Market Map: Structural Differentiation

Capability	PRISM	OpenHands	AutoGen	LangGraph	CrewAI
Multi-Model Parallel Fan-Out	✓ Yes: Native Tri-Model	✗ No: Sequential/Single	✗ No: Sequential/Single	✗ No: Sequential/Single	✗ No: Sequential/Single
Cryptographic Governance	✓ Yes: Boot-checked PAD	✗ No: Prompt-bound	✗ No: Prompt-bound	✗ No: Prompt-bound	✗ No: Prompt-bound
Instance Isolation Enforcement	✓ Yes: 3-Tier validation	N/A	N/A	N/A	N/A
Structured XML Aggregation	✓ Yes: Fusion	✗ No: Concatenation	✗ No: Concatenation	✗ No: Concatenation	✗ No: Concatenation
Full Audit/CAC Identity Chain	✓ Yes: Immutable SQLite	Warning: Partial	Warning: Partial	Warning: Partial	Warning: Partial

Only PRISM offers governed, multi-model generation.

Dual-Licensing Commercialization Strategy

PRISM Open Core (Apache-2.0)

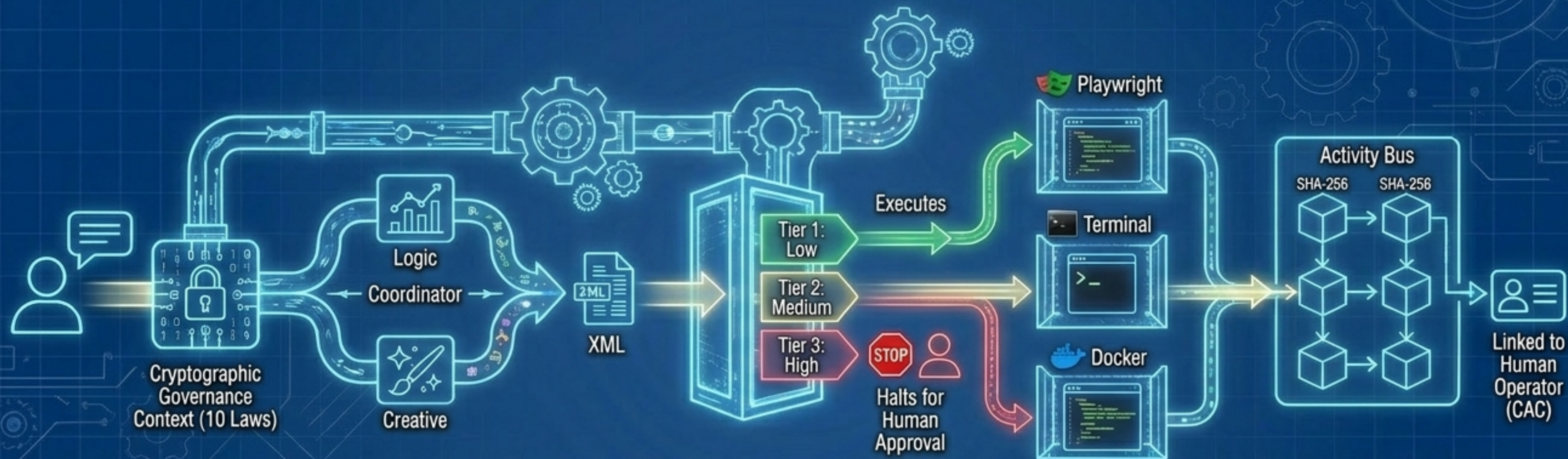
- **Audience:** Individual developers, researchers
- **Offering:** Free. Core orchestration, Spectrum Refraction, local workspace
- **Strategy:** Top-of-funnel customer acquisition, massive community adoption, standardizing the runtime

Enterprise
Upgrade Path

PRISM Business / Cloud (Commercial)

- **Audience:** Mid-market & Fortune 500 enterprises
- **Offering:** \$150-\$300 / user / month (Business) or Usage-Based (Cloud)
- **Features:** SAML/OIDC SSO, SCIM v2 provisioning, SOC2 compliance telemetry, multi-tenant isolation, managed hosting

Agents-as-a-Service (AaaS): The Unified Flow



The Mandate

User issues a prompt. System injects cryptographic governance context (10 Laws) before any processing begins.

The Cognition

The task is split into Logic and Creative pathways, evaluated concurrently, and fused by the Coordinator via XML.

The Gateway

The proposed action is classified via the 3-Tier Policy Engine. Low risk executes; high risk halts for human approval.

The Execution

Playwright, Terminal, or Docker securely enacts the approved command in an isolated sandbox.

The Record

The action is permanently recorded to the Activity Bus with SHA-256 event hashing and linked to the human operator (CAC).

The old paradigm: “Trust our model to be safe.”

The new paradigm: ‘Verify our architecture is secure.’

PRISM is the foundational operating system demonstrating that when governance is built into the architecture, human oversight and AI autonomy become complementary forces, not opposing ones.